Expressions & Equations

Name_____ Period____

The = sign is a border between friendly nations.

Both sides must hold the same amount to keep the peace.

Each box hides an unknown amount. **Our job: how many are in each box?**

Check: put your answer back in each box. Does the peace hold?

Model	□ ++ | −−−
Equation	x + 2 = −3
Record	x = −5

Evaluate Expressions

When we know what's in the box:
fill every box, then compute.



3x − 4, for x = −2

① = 3(−2) − 4 *fill each box*

② = −6 − 4 *3 groups of −2*

③ = −10 *Party! (same signs)*

Example 1



2x + 5, for x = −3

① = 2(−3) + 5 *fill each box*

② = −6 + 5 *Battle! cancel 5 pairs*

③ = −1

Example 2



6 − x, for x = −4

① = 6 −(−4) *fill the box*

② = 6 + 4 *opposite box flips −4 to +4*

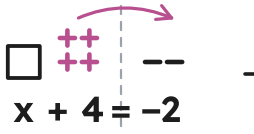
③ = 10

ONE-STEP EQUATIONS

Moving Counters

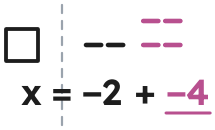
Only things you can draw can cross the border.
When they cross, they **switch teams**.

A · Switch sides



x + 4 = −2

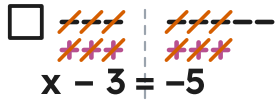
Check: −6 + 4 = −2 ✓



x = −2 + −4

x = −6

B · Add the same to both sides



x − 3 = −5

add +3 to both sides

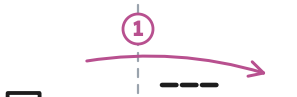
cancel pairs

+ (+3) + (+3)

x = −2

Check: −2 − 3 = −5 ✓

Example 1



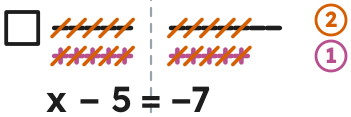
x + 3 = −6

① x = −6 + −3 *+3 crosses, becomes −3*

② x = −9

Check: −9 + 3 = −6 ✓

Example 2



x − 5 = −7

① + (+5) + (+5) *add +5 to both sides*

② x = −2 *cancel the pairs*

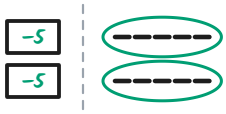
Check: −2 − 5 = −7 ✓

ONE-STEP EQUATIONS

Sharing Among Boxes

The number in front of x tells **how many boxes**.
Share the other side evenly: **one share per box**.

A



2x = −10

÷ 2 ÷ 2

x = −5

−5 hides in each box

B



−x = 5

Opposite box: what's inside gets flipped.

x = −5

or switch both: −5 = x

Example 1



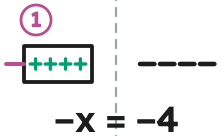
3x = −12

① ÷ 3 ÷ 3

② x = −4

share −12 into 3 boxes

Example 2



−x = −4

① x = 4

opposite box: +4 inside flips to −4

Check: −(4) = −4 ✓

SOLVING EQUATIONS

Fraction Equations

$\frac{2}{3}x$: split the box into 3 equal parts, take 2

Remember: $\frac{x}{3}$ means $\frac{1}{3}x$



$\frac{2}{3}x = -6$

① ÷ 2 ÷ 2 — 2 parts = −6, so 1 part = −3

$\frac{1}{3}x = -3$

② × 3 × 3 — 3 parts make a whole box

$x = \boxed{-9}$

Check: (2/3)(−9) = −6 ✓

Example 1



$\frac{1}{3}x = -2$

① × 3 × 3 3 parts

② $x = \boxed{-6}$

1 part = −2

Example 2



$\frac{x}{5} = -3$

① × 5 × 5 5 parts

② $x = \boxed{-15}$

x/5 means 1/5 of x

Example 3



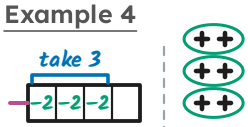
$\frac{2}{5}x = -12$

① ÷ 2 ÷ 2 1 part = −6

② × 5 × 5 5 parts

③ $x = \boxed{-30}$

Example 4



$-\frac{3x}{4} = 6$

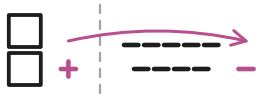
① $\frac{3}{4}x = -6$ opposite box

② $x = \boxed{-8}$ ÷3, ×4

SOLVING EQUATIONS

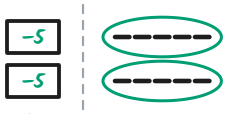
Two-Step Equations

Move the counters first,
then **share among the boxes**.



$2x + 1 = -9$

① $2x = -9 + -1$ — +1 crosses, becomes −1



share −10 into 2 boxes

$2x = -10$

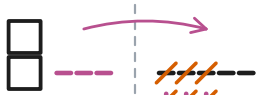
② ÷ 2 ÷ 2

$x = \boxed{-5}$ Check: 2(−5) + 1 = −9 ✓

Watch out! The minus belongs to the box:

$\boxed{5}\boxed{-x} = -3$ circle the number, box the −x

Example 1



$2x - 3 = -5$

① $2x = -5 + 3$
−3 crosses, becomes +3

② $x = \boxed{-1}$
share −2 into 2 boxes

Check: 2(−1) − 3 = −5 ✓

Example 2



$5 - x = -3$

① $-x = -3 + -5$
+5 crosses, becomes −5

② $x = \boxed{8}$
opposite box: −8 flips to 8

Check: 5 − 8 = −3 ✓

Example 3



$-3x - 1 = 8$

① $-3x = 8 + 1$
−1 crosses, becomes +1

② $x = \boxed{-3}$
share 9: each opposite box = +3

Check: −3(−3) − 1 = 8 ✓

Example 4



$\frac{1}{2}x - 4 = -9$

① $\frac{1}{2}x = -9 + 4$
−4 crosses, becomes +4

② $x = \boxed{-10}$
half box = −5, whole = −10

Check: (1/2)(−10) − 4 = −9 ✓

SOLVING EQUATIONS

Distribute, then Solve

Open the groups first, then solve.

$3 - (x - 4) = -1$

① $3 + -1(x - 4) = -1$ — subtract = add the opposite

② $3 - x + 4 = -1$ — distribute −1

③ $7 - x = -1$ — combine 3 + 4

④ $-x = -1 + -7$ — +7 crosses, becomes −7

⑤ $x = \boxed{8}$ — opposite box: −8 flips to 8

Check: 3 − (8 − 4) = −1 ✓

Example 1

$2(3x - 1) + 5 = -9$

① $6x - 2 + 5 = -9$
distribute the 2

② $6x + 3 = -9$
combine −2 + 5

③ $6x = -9 + -3$
+3 crosses, becomes −3

④ $x = \boxed{-2}$
share −12 into 6 boxes

Check: 2(3(−2) − 1) + 5 = −9 ✓

Example 2

$2x - 3(x - 1) = 6$

① $2x + -3(x - 1) = 6$
subtract = add the opposite

② $2x - 3x + 3 = 6$
distribute −3

③ $-x + 3 = 6$
combine 2x − 3x

④ $-x = 6 + -3$
+3 crosses, becomes −3

⑤ $x = \boxed{-3}$
opposite box: 3 flips to −3

Check: 2(−3) − 3(−3 − 1) = 6 ✓